F78PA Probability & Statistics A

Tutorial 1 – Solutions

1. (a) The sample space Ω consists of all ordered triples of the form (i, j, k) where i, j, k are integers between 1 and 6.

$$\Omega = \{(1,1,1), (1,1,2), (1,1,3), ..., (6,6,6)\}$$

Note: (i, j, k) corresponds to the event that the first die shows i, the second die shows j, and the third die shows k.

(b) Let $E = \{\text{all 3 dice are the same}\} = \{(1,1,1), (2,2,2), (3,3,3), (4,4,4), (5,5,5), (6,6,6)\}$. So,

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{6}{6^3} = \frac{1}{36}.$$

(c) Let $A = \{\text{all three dice are different}\}$

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{654}{6^3} = \frac{5}{9}.$$

(d) Let $F = \{\text{at least one die shows an even number}\}$. Use the Law of Complements,

$$\mathbb{P}(F) = 1 - \mathbb{P}(F^c) = 1 - \mathbb{P}(\text{all dice are odd}) = 1 - \frac{3 \times 3 \times 3}{6^3} = 1 - \frac{1}{8} = \frac{7}{8}.$$

2. (a) The midterm questions will be a subset of size 4 from 9 questions. Then,

$$|\Omega| = \binom{9}{4} = 126.$$

(b) i. Let $E = \{\text{You can solve all 4 problems}\}$

$$\mathbb{P}(E) = \frac{\binom{6}{4}}{\binom{9}{4}} = \frac{15}{126} = \frac{5}{42}.$$

ii. Let $F = \{\text{1 question from the 3 you can't solve, and 3 from the 6 you can solve}\}$.

$$\mathbb{P}(F) = \frac{\binom{3}{1} \times \binom{6}{3}}{\binom{9}{4}} = \frac{60}{126} = \frac{10}{21}.$$

iii. Let $A = \{\text{at least 2 questions come from the 6 that you can solve}\}$. Use the Law of Complements,

$$\mathbb{P}(A) = 1 - \mathbb{P}(A^c) = 1 - \mathbb{P}(B) - \mathbb{P}(C)$$

$B = \{\text{You can solve only 1 problem}\}$, where $|B| = \binom{6}{1} \times \binom{3}{3} = 6$.
$C = \{\text{You can't solve any problem}\}$, where $C = \emptyset$. (Since if 4 problems are chosen, at least one of them must be from the 6 that you can solve) So, $\mathbb{P}(A) = 1 - \mathbb{P}(B) - \mathbb{P}(C) = 1 - \frac{6}{\binom{9}{4}} - 0 = \frac{120}{126} = \frac{20}{21}$.

3. (a) From $P(k) = \frac{3}{4^k}$ for $k = 1, 2, 3, ...$ All of these values are non-negative, so it is enough to check that $P(1) + P(2) + P(3) + ... = 1$.

$$P(1)+P(2)+P(3)+... = \frac{3}{4}+\frac{3}{4^2}+\frac{3}{4^3}+... = \frac{3}{4}\left(1+\frac{1}{4}+\frac{1}{4^2}+...\right) = \frac{3}{4}\times\frac{1}{1-\frac{1}{4}} = 1.$$

So, $P(k) = \frac{3}{4^k}$ is a probability function.

(b) Let $A = \{\text{outcome is greater than 3}\}$.

$$\mathbb{P}(A) = 1 - \mathbb{P}(A^c) = 1 - P(1) - P(2) - P(3) = 1 - \frac{3}{4} - \frac{3}{4^2} - \frac{3}{4^3} = 1 - \frac{48+12+3}{64} = \frac{1}{64}.$$

4. Suppose $\Omega = \{0, 1, 2, 3, \ldots\}$, and $P(k) = c\frac{4^k}{k!}$. We need to find the value of c such that $P(0) + P(1) + P(2) + \ldots = 1$. That is,

$$1 = c\sum_{k=0}^{\infty} \frac{4^k}{k!}.$$

Note that

$$\sum_{k=0}^{\infty} \frac{4^k}{k!} = e^4.$$

Therefore we must set $c = e^{-4}$.

5. The event that it is either too wet or too windy is $A \cup B$ and we have

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) = 0.35 + 0.30 - 0.20 = 0.45.$$

Hence by the Law of Complements, the probability that the match can go ahead is $\mathbb{P}((A \cup B)^c) = 1 - 0.45 = 0.55$.

6. Either the event A or the event B *must* happen, and so $\mathbb{P}(A \cup B) = 1$. The event that the device works is $A \cap B$. Now using the inclusion-exclusion rule,

$$\begin{aligned} \mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \\ 1 &= 0.6 + 0.75 - \mathbb{P}(A \cap B) \\ \Rightarrow \quad \mathbb{P}(A \cap B) &= 0.35 \end{aligned}$$

The probability that the device works is 0.35.

7. (a) Show that $\mathbb{P}(E) = 1 - \mathbb{P}(E^c)$
First notice that Ω is the disjoint union of E and E^c, so by property (III),

$$\mathbb{P}(\Omega) = \mathbb{P}(E \cup E^c) = \mathbb{P}(E) + \mathbb{P}(E^c)$$

Now by property (II), $\mathbb{P}(\Omega) = 1$, and so $\mathbb{P}(E) = 1 - \mathbb{P}(E^c)$

(b) Show that $P(E \cap F^c) = P(E) - P(E \cap F)$
First, notice that $E = (E \cap F) \cup (E \cap F^c)$, and that this is a disjoint union.
By property (III), $\mathbb{P}(E) = \mathbb{P}(E \cap F) + \mathbb{P}(E \cap F^c)$, and re-arranging this equation gives the required result.

(c) Show that $P(E \cup F) = P(E) + P(F) - P(E \cap F)$
First, notice that $E \cup F = E \cup (F \cap E^c)$, and that this is a disjoint union (draw a Venn diagram to see this).
By property (III), $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F \cap E^c)$
Now from part (b) above we know that $P(F \cap E^c) = P(F) - P(E \cap F)$, so substituting this in, we obtain $\mathbb{P}(E \cup F) = \mathbb{P}(E) + P(F) - P(E \cap F)$, as required.

8. Start by defining some notation for events:

$$\begin{aligned} J &= \{\text{worker gets the job}\} \\ S &= \{\text{supervisor writes a strong reference}\} \\ M &= \{\text{supervisor writes a moderately strong reference}\} \\ W &= \{\text{supervisor writes a weak reference}\} \end{aligned}$$

The question gives us the following information:

$$\begin{gathered} \mathbb{P}(J \mid S) = 0.8 \quad \mathbb{P}(J \mid M) = 0.4 \quad \mathbb{P}(J \mid W) = 0.1 \\ \mathbb{P}(S) = 0.7 \qquad \mathbb{P}(M) = 0.2 \qquad \mathbb{P}(W) = 0.1 \end{gathered}$$

(a) Law of Total Probability:

$$\begin{aligned}\mathbb{P}(J) &= \mathbb{P}(S)\mathbb{P}(J \mid S) + \mathbb{P}(M)\mathbb{P}(J \mid M) + \mathbb{P}(W)\mathbb{P}(J \mid W) \\ &= 0.7 \times 0.8 + 0.2 \times 0.4 + 0.1 \times 0.1 \\ &= 0.65\end{aligned}$$

The probability that the worker gets the job is 0.65.

(b) Using the definition of conditional probability,

i. $$\mathbb{P}(S \mid J) = \frac{\mathbb{P}(S \cap J)}{\mathbb{P}(J)} = \frac{\mathbb{P}(S)\mathbb{P}(J \mid S)}{\mathbb{P}(J)} = \frac{0.7 \times 0.8}{0.65} = \frac{56}{65} \approx 0.8615$$

ii. $$\mathbb{P}(M \mid J) = \frac{\mathbb{P}(M \cap J)}{\mathbb{P}(J)} = \frac{\mathbb{P}(M)\mathbb{P}(J \mid M)}{\mathbb{P}(J)} = \frac{0.2 \times 0.4}{0.65} = \frac{8}{65} \approx 0.1231$$

iii. $$\mathbb{P}(W \mid J) = \frac{\mathbb{P}(W \cap J)}{\mathbb{P}(J)} = \frac{\mathbb{P}(W)\mathbb{P}(J \mid W)}{\mathbb{P}(J)} = \frac{0.1 \times 0.1}{0.65} = \frac{1}{65} \approx 0.0154$$

(c) First, note that by the Law of Complements,

$$\mathbb{P}(\text{worker does not get the job}) = \mathbb{P}(J^c) = 1 - \mathbb{P}(J) = 1 - 0.65 = 0.35$$

Similarly, using the Law of Complements for conditional probabilities, note that

$$\begin{aligned}\mathbb{P}(J^c \mid S) &= 1 - \mathbb{P}(J \mid S) = 1 - 0.8 = 0.2 \\ \mathbb{P}(J^c \mid M) &= 1 - \mathbb{P}(J \mid M) = 1 - 0.4 = 0.6 \\ \mathbb{P}(J^c \mid W) &= 1 - \mathbb{P}(J \mid W) = 1 - 0.1 = 0.9\end{aligned}$$

Then

$$\mathbb{P}(S \mid J^c) = \frac{\mathbb{P}(S \cap J^c)}{\mathbb{P}(J^c)} = \frac{\mathbb{P}(S)\mathbb{P}(J^c \mid S)}{\mathbb{P}(J^c)} = \frac{0.7 \times 0.2}{0.35} = \frac{2}{5} = 0.4$$

$$\mathbb{P}(M \mid J^c) = \frac{\mathbb{P}(M \cap J^c)}{\mathbb{P}(J^c)} = \frac{\mathbb{P}(M)\mathbb{P}(J^c \mid M)}{\mathbb{P}(J^c)} = \frac{0.2 \times 0.6}{0.35} = \frac{12}{35} \approx 0.3429$$

$$\mathbb{P}(W \mid J^c) = \frac{\mathbb{P}(W \cap J^c)}{\mathbb{P}(J^c)} = \frac{\mathbb{P}(W)\mathbb{P}(J^c \mid W)}{\mathbb{P}(J^c)} = \frac{0.1 \times 0.9}{0.35} = \frac{9}{35} \approx 0.2571$$

9. First define notation for events:

$$E_i = \{\text{trainee passes } i\text{th exam}\} \text{ for } i = 1, 2, 3$$

We are told that

$$\mathbb{P}(E_1) = 0.9 \quad \mathbb{P}(E_2 \mid E_1) = 0.8 \quad \mathbb{P}(E_3 \mid E_1 \cap E_2) = 0.7$$

(a) By the 'chain rule' for three events,

$$\begin{aligned}\mathbb{P}(E_1 \cap E_2 \cap E_3) &= \mathbb{P}(E_1)\mathbb{P}(E_2 \mid E_1)\mathbb{P}(E_3 \mid E_1 \cap E_2) \\ &= 0.9 \times 0.8 \times 0.7 = 0.504\end{aligned}$$

The probability that he passess all three exams is 0.504.

(b) Define the event

$$N = \{\text{trainee did not pass all three exams}\} = (E_1 \cap E_2 \cap E_3)^c$$

and note that the event 'trainee failed the first exam' is E_1^c.
Using the definition of conditional probability, the required probability is

$$\mathbb{P}(E_1^c \mid N) = \frac{\mathbb{P}(E_1^c \cap N)}{\mathbb{P}(N)}$$

But if the trainee fails the first exam, then he certainly does not pass all three exams; that is, $E_1^c \subseteq N$. Hence $E_1^c \cap N = E_1^c$, and so

$$\begin{aligned}
\mathbb{P}(E_1^c \mid N) = \frac{\mathbb{P}(E_1^c \cap N)}{\mathbb{P}(N)} &= \frac{\mathbb{P}(E_1^c)}{\mathbb{P}(N)} \\
&= \frac{1-\mathbb{P}(E_1)}{1-\mathbb{P}(N^c)} \quad \text{(Law of Complements)} \\
&= \frac{1-\mathbb{P}(E_1)}{1-\mathbb{P}(E_1 \cap E_2 \cap E_3)} \\
&= \frac{1-0.9}{1-0.504} = \frac{25}{124} \approx 0.2016
\end{aligned}$$

Exam Style Question

(a) An outcome can be described as a pair (i, j) where i rep resents the number of dots appearing on one die and j represents the number of dots appearing on the other die. The sample space, which is the set of all possible outcomes, is then given as

$$\Omega = \{(a, b) : a, b \in \{1, \ldots, 6\}\}.$$

[1 mark]

(b) Let $A \subseteq \Omega$ consist of outcomes (i, j) where $i + j = 7$, i.e.

$$A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}.$$

Since the probability of each outcome is equal we have

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{6}{36} = \frac{1}{6}.$$

[2 marks]

(c) Let $C \subseteq \Omega$ consist of outcomes (i, j) where $i + j > 12$. We know however that $i, j \leq 6$ so there are no such outcomes, i.e. $C = \emptyset$. Hence, the probability of this event is zero irrespective of whether the die are fair or not. [2 marks]

(d) The probability of an event $E \in \mathcal{F}$ occurring given that an event $F \in \mathcal{F}$ occurred with $\mathbb{P}(F) > 0$ is given by

$$\mathbb{P}(E|F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}.$$

(1 mark)

(e) Let $E \subseteq \Omega$ be the event that exactly one dice shows a 6 then

$$A \cap E = \{(1, 6), (6, 1)\}$$

and $\mathbb{P}(A \cap E) = 2/36 = 1/18$, hence

$$\mathbb{P}(E|A) = \frac{\mathbb{P}(A \cap E)}{\mathbb{P}(A)} = \frac{\frac{2}{36}}{\frac{6}{36}} = \frac{1}{3}.$$

(1 marks)